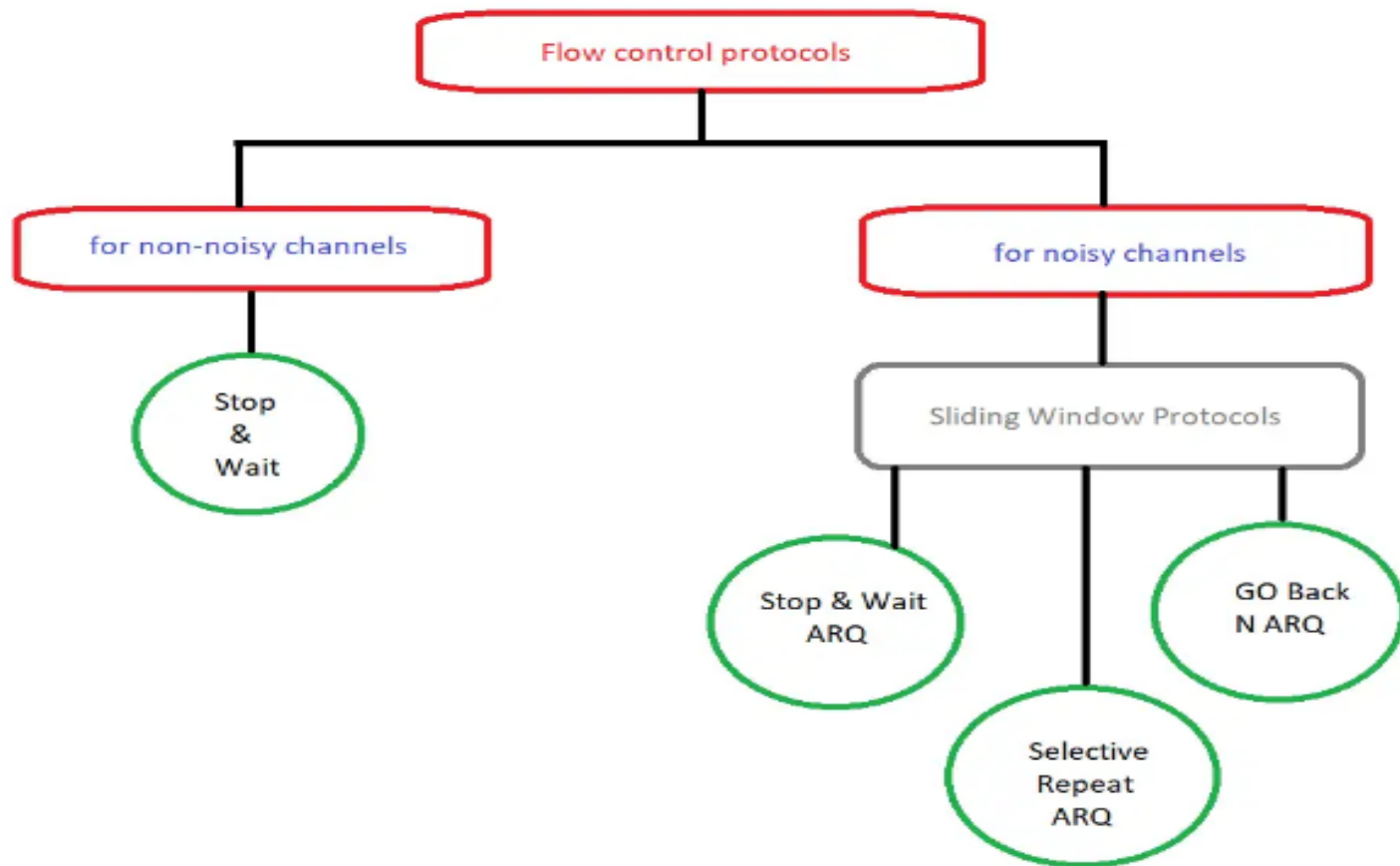
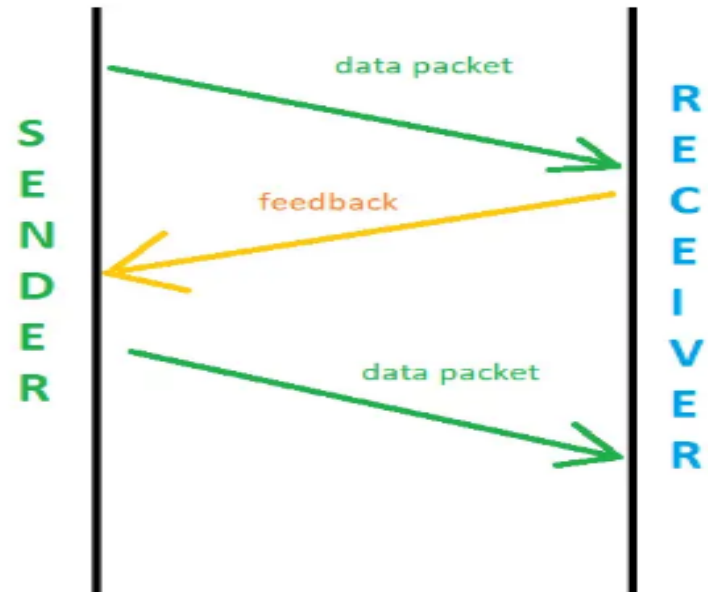


FLOW CONTROL

- Flow control in computer networks is defined as the process of managing the rate of data transmission between two systems(nodes), this mechanism ensures that the rate of data (transmitted by the sender) is within the receiving capacity of the receiver node.





Stop & Wait Protocol

Flow Control

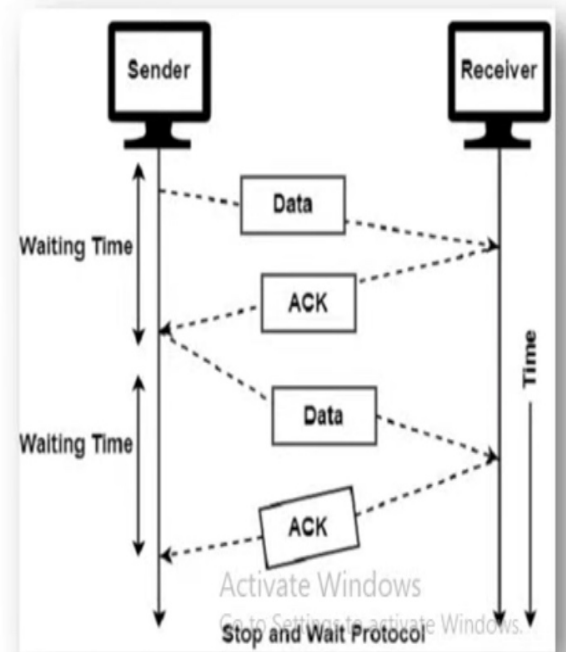
- Flow control coordinates the amount of data that can be sent before receiving acknowledgement
- It is one of the most important functions of data link layer.
- Flow control is a set of procedures that tells the sender how much data it can transmit before it must wait for an acknowledgement from the receiver.
- Receiver has a limited speed at which it can process incoming data and a limited amount of memory in which to store incoming data.
- Receiver must inform the sender before the limits are reached and request that the transmitter to send fewer frames or stop temporarily.
- Since the rate of processing is often slower than the rate of transmission, receiver has a block of memory (buffer) for storing incoming data until they are processed.

Flow Control Mechanisms

- Stop-and-Wait
- Sliding Window Protocol
 - Go-Back-N ARQ
 - Selective-Repeat ARQ

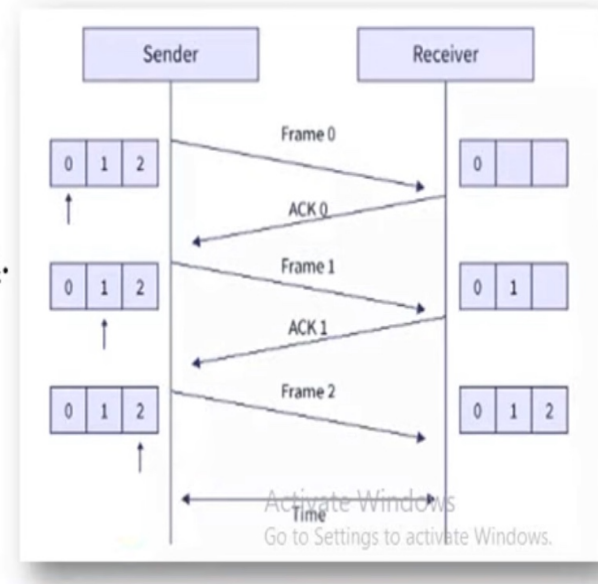
About Stop & Wait Protocol

- It used in connection oriented communication.
- It act as error & flow control protocol
- It used in Data Link Layer & Transport Layer.
- It is one of the type of Sliding Window Protocol.
- It uses a link between sender and receiver as a half-duplex link



Working of Stop & Wait Protocol

1. Sender send data packets to the receiver.
 2. Receiver receive the packet & sends acknowledgment to the sender i.e. Received.
 3. Once the acknowledgment is received, the sender sends the next packet.
 4. This process continues until all the packet are not sent.
- Sender will not send second packet without receiving the acknowledgment of the first packet.
 - Manage the flow between sender & receiver communication.



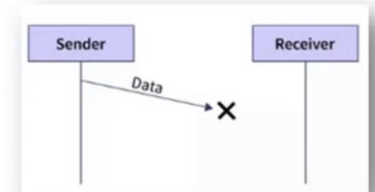
Advantages of Stop & Wait Protocol

1. **Simplicity:** It is easy to implement & provide basic communication system.
2. **Accuracy:** Sender sends the next packet to receiver only when the acknowledgment of the previous packet is received. So there is less chance of the packet being lost.
3. **Flow Control:** It ensures that information is received in the order it was sent.
4. **Error Control:** If the sender doesn't receive an acknowledgment within a timeout period, it will retransmit the frame.
5. **Efficiency:** The protocol is useful in LANs, and its efficiency increases as the size of the data packets increases. Its efficiency decreases as the distance between the sender and receiver increases.

Disadvantages of Stop & Wait Protocol

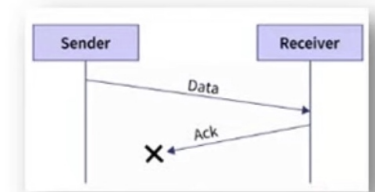
1. Lost Data:

- Suppose sender sends data and the data is lost.
- Receiver is waiting for the data for a long time.
- Data is not received by receiver, so it does not send any acknowledgment.
- Since, Sender does not receive any acknowledgment so it will not send the next packet.



2. Lost of Acknowledgement:

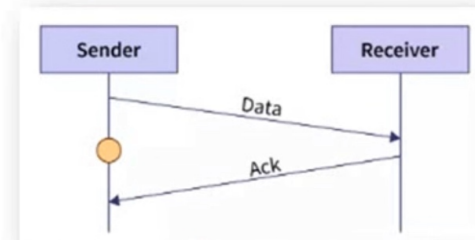
- Sender sends data and it has also been received by the receiver.
- On receiving the packet, receiver sends acknowledgment.
- In this case, the acknowledgment is lost in network, sender not receive acknowledgment.
- Sender didn't send next packet & receiver get in waiting state.



Disadvantages of Stop & Wait Protocol

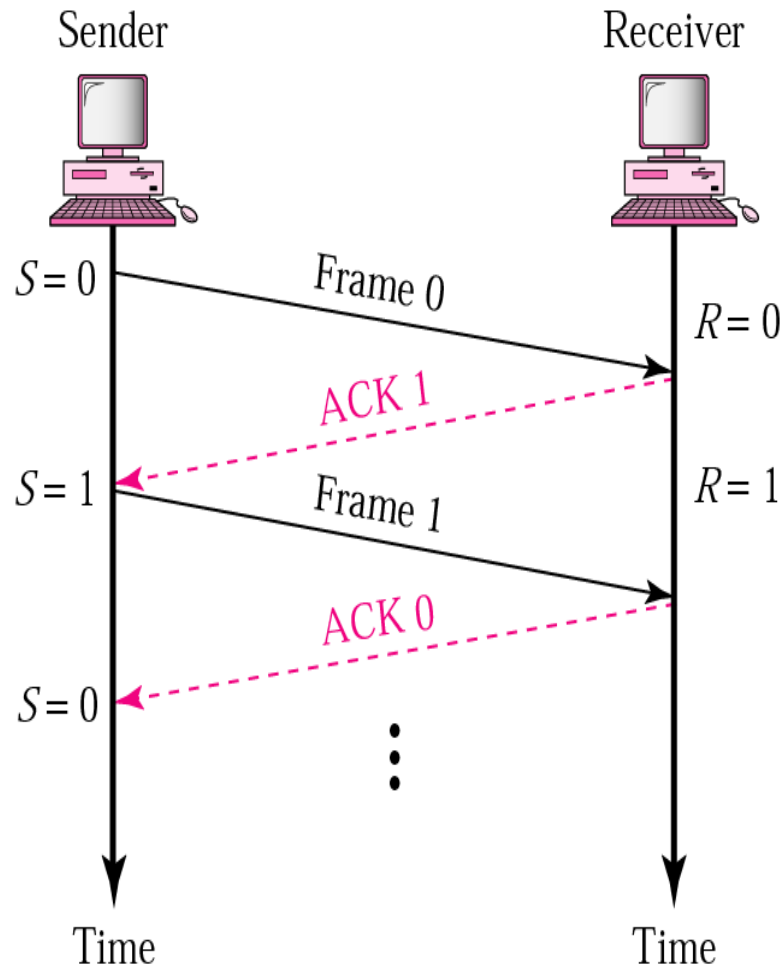
3. Delayed data or acknowledgment:

- Suppose the sender sends the data and it has also been received by the receiver.
- The receiver then sends the acknowledgment but the acknowledgment is received after the timeout period on the sender's side.
- As the acknowledgment is received late, so acknowledgment can be wrongly considered as the acknowledgment.
- Sender retransmit the packets.



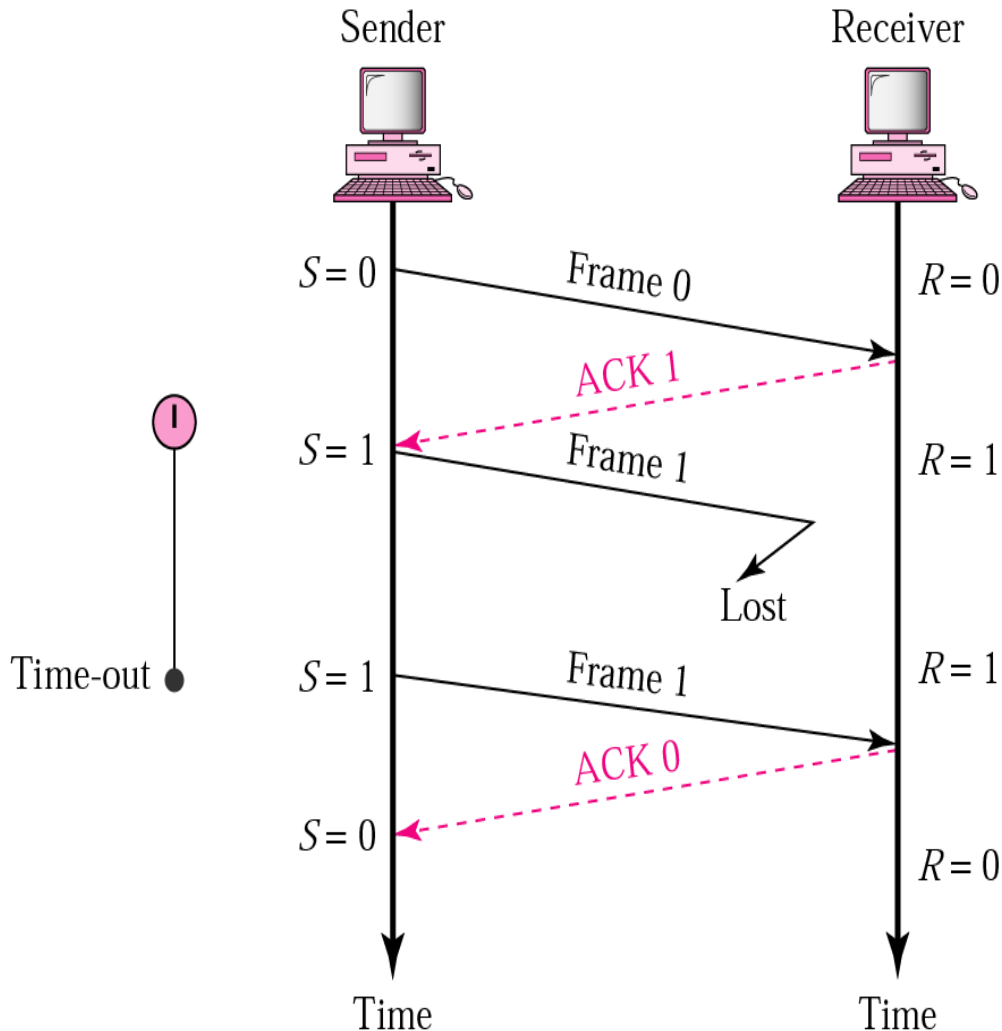
Activate Windows
Go to Settings to activate Windows.

Stop-and-Wait



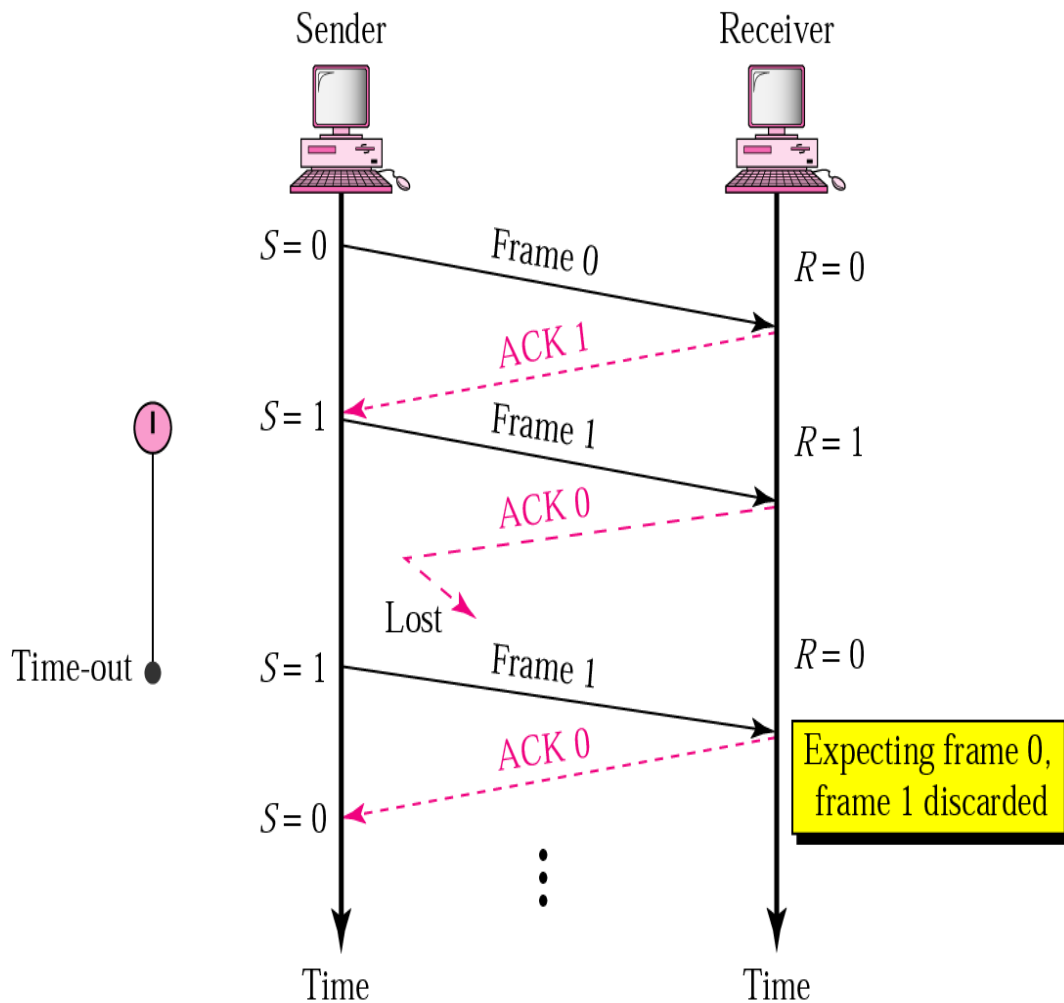
- Sender keeps a copy of the last frame until it receives an acknowledgement.
- For identification, both data frames and acknowledgements (ACK) frames are numbered alternatively 0 and 1.
- Sender has a control variable (S) that holds the number of the recently sent frame. (0 or 1)
- Receiver has a control variable R that holds the number of the next frame expected (0 or 1).
- Sender starts a timer when it sends a frame. If an ACK is not received within a allocated time period, the sender assumes that the frame was lost or damaged and resends it
- Receiver send only positive ACK if the frame is intact.
- ACK number always defines the

Stop-and-Wait ARQ, lost frame



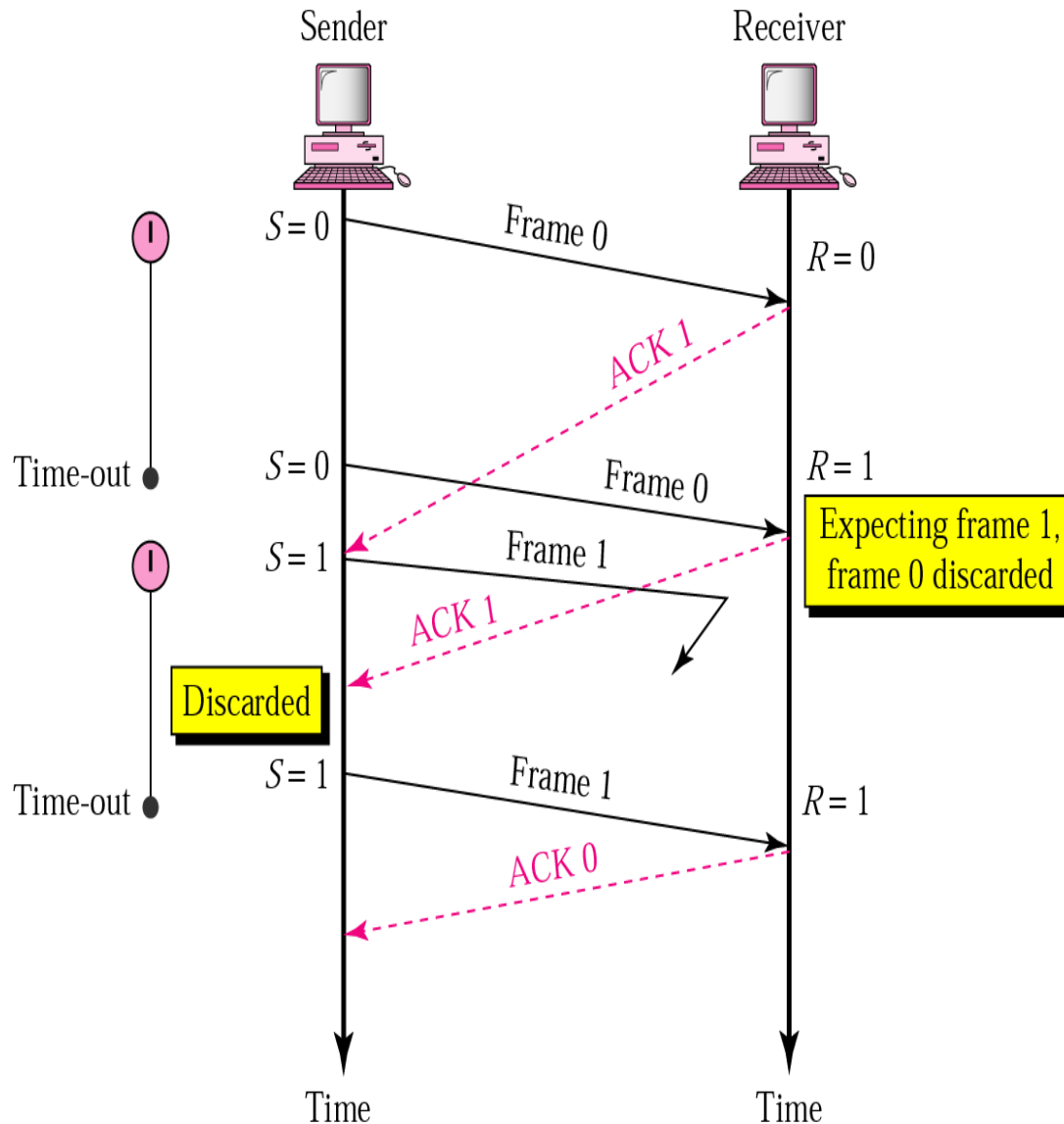
- When a receiver receives a damaged frame, it discards it and keeps its value of R .
- After the timer at the sender expires, another copy of frame 1 is sent.

stop-and-wait, lost ACK frame



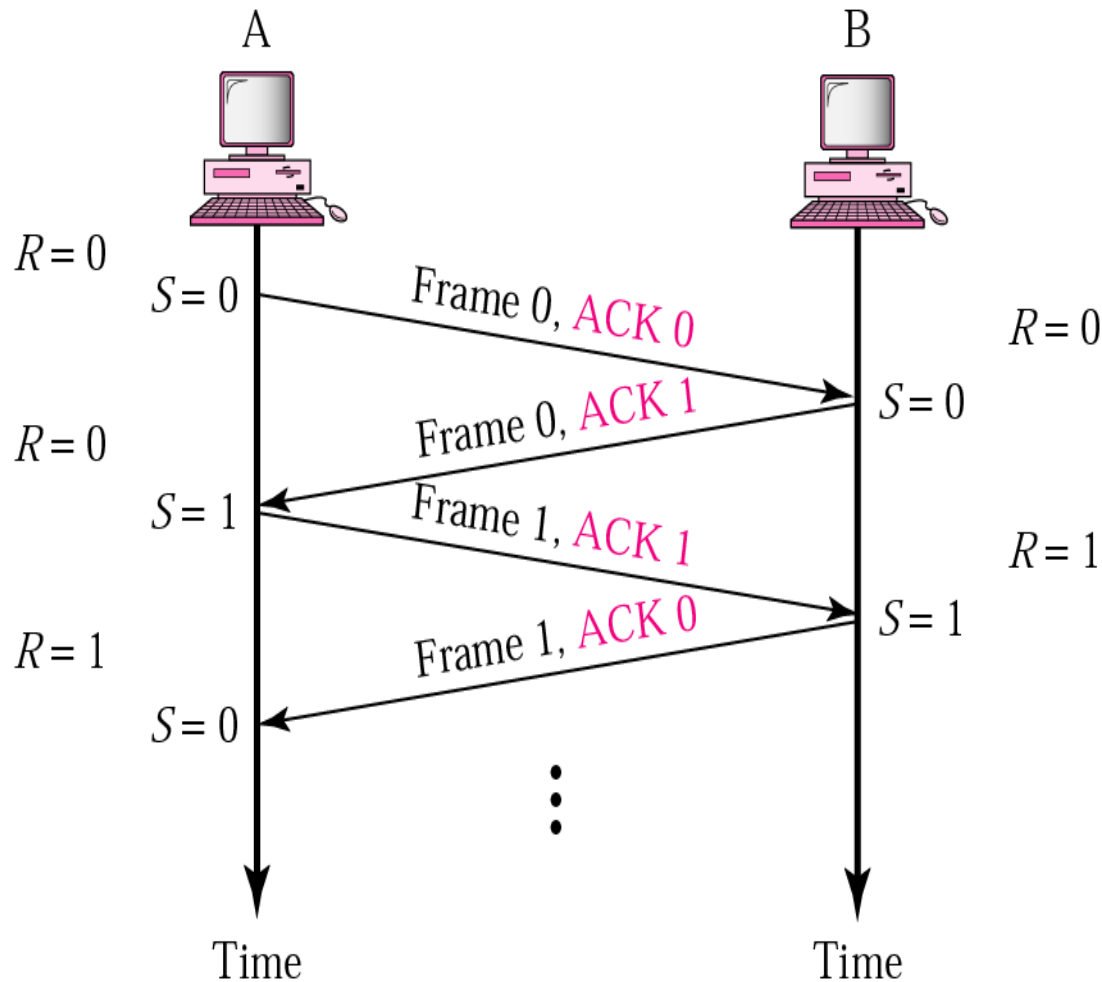
- If the sender receives a damaged ACK, it discards it.
- When the timer of the sender expires, the sender retransmits frame 1.
- Receiver has already received frame 1 and expecting to receive frame 0 ($R=0$). Therefore it discards the second copy of frame 1.

stop-and-wait, delayed ACK frame



- The ACK can be delayed at the receiver or due to some problem
- It is received after the timer for frame 0 has expired.
- Sender retransmitted a copy of frame 0. However, $R=1$ means receiver expects to see frame 1. Receiver discards the duplicate frame 0.
- Sender receives 2 ACKs, it discards the second ACK.

Piggybacking



- A method to combine a data frame with ACK.
- Station A and B both have data to send.
- Instead of sending separately, station A sends a data frame that includes an ACK.
- Station B does the same thing.
- Piggybacking saves bandwidth.

Disadvantage of Stop-and-Wait

- In stop-and-wait, at any point in time, there is only one frame that is sent and waiting to be acknowledged.
- This is not a good use of transmission medium.
- To improve efficiency, multiple frames should be in transition while waiting for ACK.
- Two protocols use the above concept,
 - **Go-Back-N ARQ**
 - **Selective Repeat ARQ**

About Sliding Window Protocol

- These are data link layer protocols for reliable and sequential delivery of data frames.
- It is also used in Transmission Control Protocol (TCP).
- This technique is for sending multiple frames at a time to the receiver.
- Each frame has sent from the sequence number.
- The sequence numbers are used to find the missing data in the receiver end.
- The main purpose is to avoid duplicate data, so it uses the sequence number.

➤Types of Sliding Window Protocol:

1. Go-Back-N ARQ
2. Selective Repeat ARQ



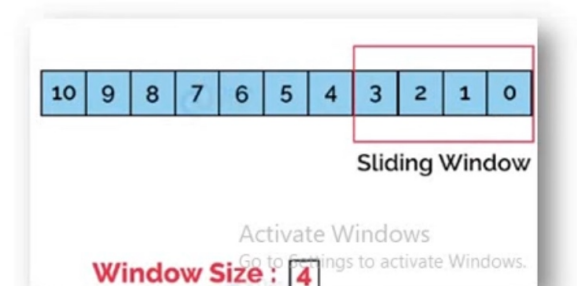
SLIDING WINDOW

- In the **stop-and-wait protocol**, the sender can send only one frame at a time and cannot send the next frame without receiving the acknowledgment of the previously sent frame, whereas, in the case of **sliding window protocol**, the multiple frames can be sent at a time.

- Sliding window protocols are data link layer protocols for reliable and sequential delivery of data frames.
- The sliding window is also used in Transmission Control Protocol.
- **Types of Sliding Window Protocol**
 - 1. Go-Back-N ARQ**
 - 2. Selective Repeat ARQ**

Go-Back-N ARQ Protocol

- Go Back N Automatic Repeat Request (ARQ) is a data link layer protocol.
- It is used for data flow control purposes.
- In which multiple frames are sent from sender to receiver at once.
- The sender window is a fixed-sized window 'N' that defines the number of frames that are transmitted from sender to receiver at once.
- The Receiver window in the Go Back N ARQ protocol is always of size 1.
- Receiver takes at most 1 frame at a single time.
- It reject corrupted frames & avoid out of sequence frames.



GO-BACK-N ARQ

- In Go-Back-N ARQ, **N is the sender's window size** and the size of the **receiver's window is always 1**.
- The number of frames that can be sent at a time totally depends on the size of the sender's window.
- So, we can say that 'N' is the number of frames that can be sent at a time before receiving the acknowledgment from the receiver.

- Go-Back-N ARQ uses a cumulative acknowledgment technique.
- The frames are sequentially numbered and a finite number of frames are sent. If the acknowledgment of a frame is not received within the time period, all frames starting from that frame are retransmitted.

- The size of the sending window determines the sequence number of the outbound frames.
- If the sequence number of the frames is an m -bit field, then the range of sequence numbers that can be assigned is 0 to $2^m - 1$.
- **The size of the sending window is $2^m - 1$.**
- In Go-Back-N ARQ, the size of the send window must be less than 2^m ; the size of the receiver window is always 1.
- **sequence numbers required = sender window size + receiver window size**

Go-Back-N ARQ

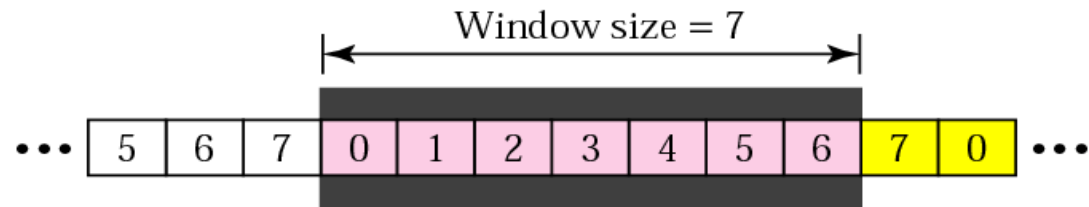
- We can send up to W frames before worrying about ACKs.
- We keep a copy of these frames until the ACKs arrive.
- This procedure requires additional features to be added to Stop-and-Wait ARQ.

Sequence Numbers

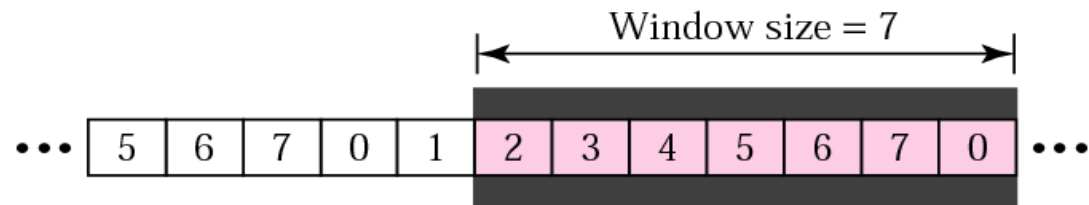
- Frames from a sender are numbered sequentially.
- We need to set a limit since we need to include the sequence number of each frame in the header.
- If the header of the frame allows m bits for sequence number, the sequence numbers range from 0 to $2^m - 1$. for $m = 3$, sequence numbers are: 1, 2, 3, 4, 5, 6, 7.
- We can repeat the sequence number.
- Sequence numbers are:
0, 1, 2, 3, 4, 5, 6, 7, 0, 1, 2, 3, 4, 5, 6, 7, 0, 1, ...

Sender Sliding Window

- At the sending site, to hold the outstanding frames until they are acknowledged, we use the concept of a window.
- The size of the window is at most $2^m - 1$ where m is the number of bits for the sequence number.
- Size of the window can be variable, e.g. TCP.
- The window slides to



a. Before sliding

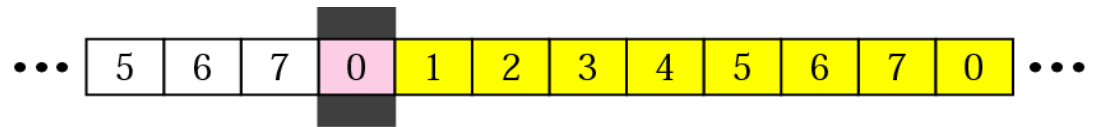


b. After sliding two frames

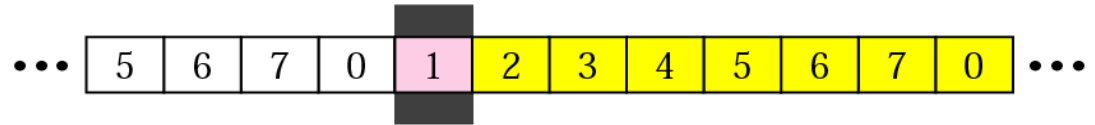
Receiver Sliding Window

- Size of the window at the receiving site is always 1 in this protocol.
- Receiver is always looking for a specific frame to arrive in a specific order.
- Any frame arriving out of order is discarded and needs to be resent.
- Receiver window slides as shown in fig. Receiver is

waiting for frame 0



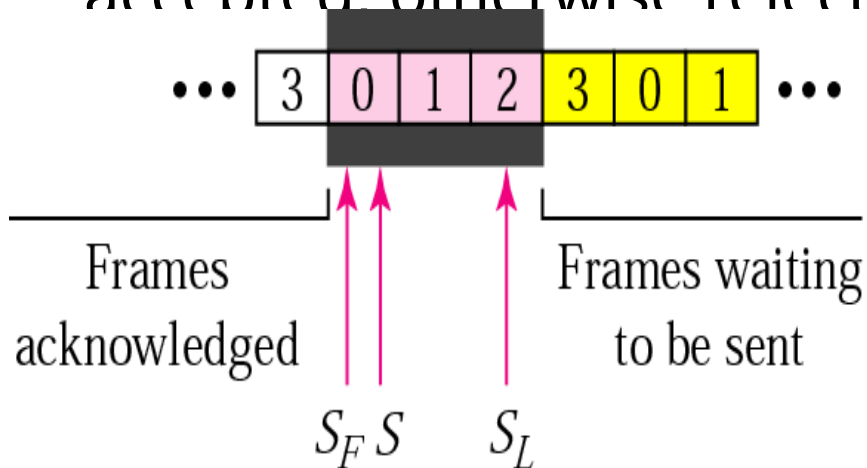
a. Before sliding



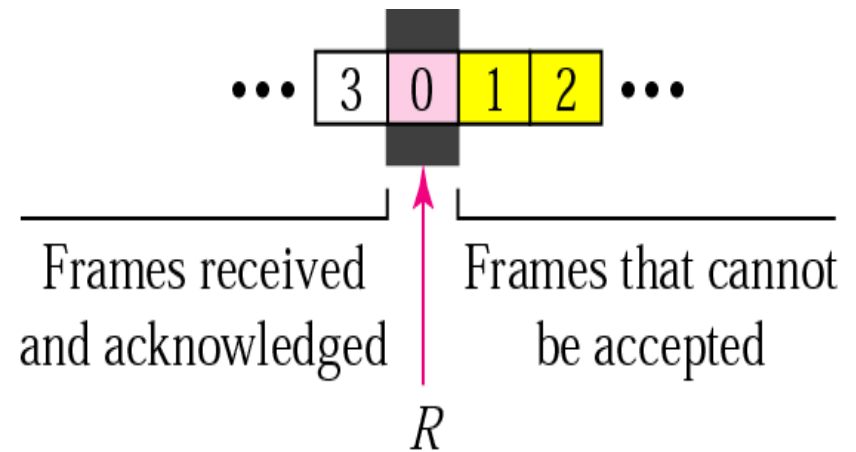
b. After sliding

Control Variables

- Sender has 3 variables: S , S_F , and S_L
- S holds the sequence number of recently sent frame
- S_F holds the sequence number of the first frame
- S_L holds the sequence number of the last frame
- Receiver only has the one variable, R , that holds the sequence number of the frame it expects to receive. If the seq. no. is the same as the value of R , the frame is accepted, otherwise rejected.



a. Sender window



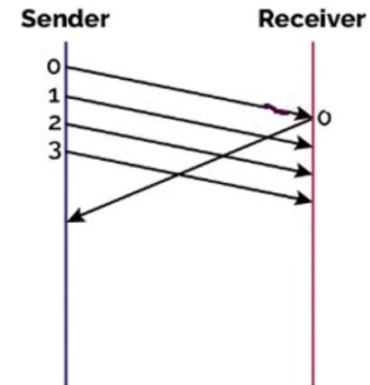
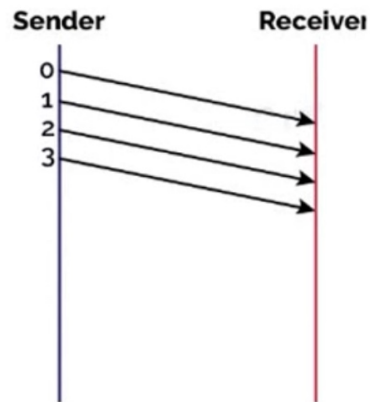
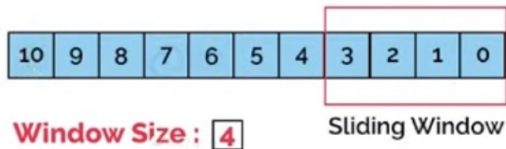
b. Receiver window

Acknowledgement

- Receiver sends positive ACK if a frame arrived safe and in order.
- If the frames are damaged/out of order, receiver is silent and discard all subsequent frames until it receives the one it is expecting.
- The silence of the receiver causes the timer of the unacknowledged frame to expire.
- Then the sender resends all frames, beginning with the one with the expired timer.
- For example, suppose the sender has sent frame 6, but the timer for frame 3 expires (i.e. frame 3 has not been acknowledged), then the sender goes back and sends frames 3, 4, 5, 6 again. Thus it is called Go-Back-N-ARQ
- The receiver does not have to acknowledge each frame received, it can send one cumulative ACK for several

Example of Go-Back-N ARQ

- **Step 1:** Sender will send first four frames to the receiver (0,1,2,3). $N = 4$.
- **Step 2:** Sender expects to get ack from 0th frame.

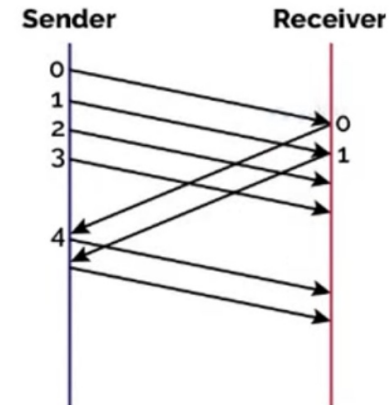
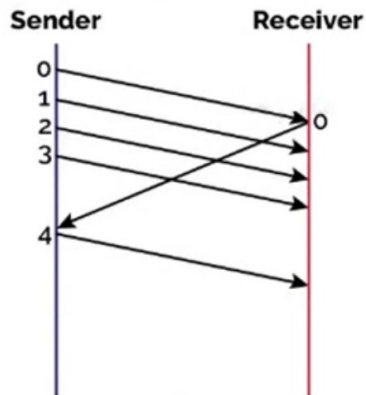
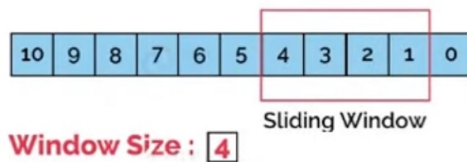


Activate Windows
Go to Settings to activate Windows.

Example of Go-Back-N ARQ

➤ **Step 3:** Sender will send following frame, which is 4. The window slides containing four frames (1,2,3,4).

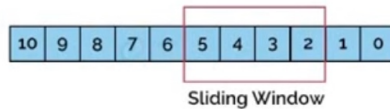
➤ **Step 4:** The acknowledgment for frame no. 1 will then be sent by the receiver.



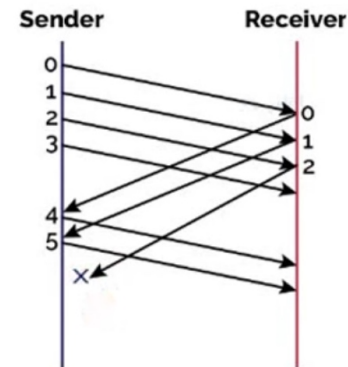
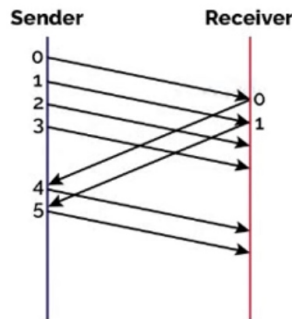
Activate Windows
Go to Settings to activate Windows.

Example of Go-Back-N ARQ

- **Step 5:** Sender will then send following frame, **which is 5**. The window slides containing four frames (2,3,4,5).
- **Step 6:** The acknowledgment for frame no. 2 will then be sent by the receiver. But acknowledgment is lost in network. Not received by the Sender.



Window Size : 4

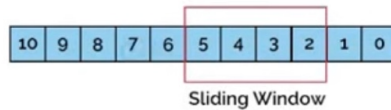


Activate Windows
Go to Settings to activate Windows.

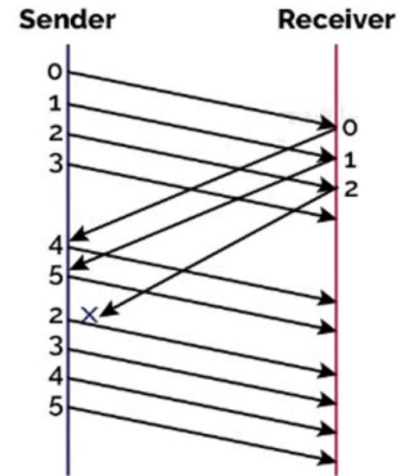
Example of Go-Back-N ARQ

➤ **Step 7:** Instead of transmitting frame number 6, the sender Go-Back to 2, which is the initial frame of the current window.

Retransmits all frames in the current window (2,3,4,5).

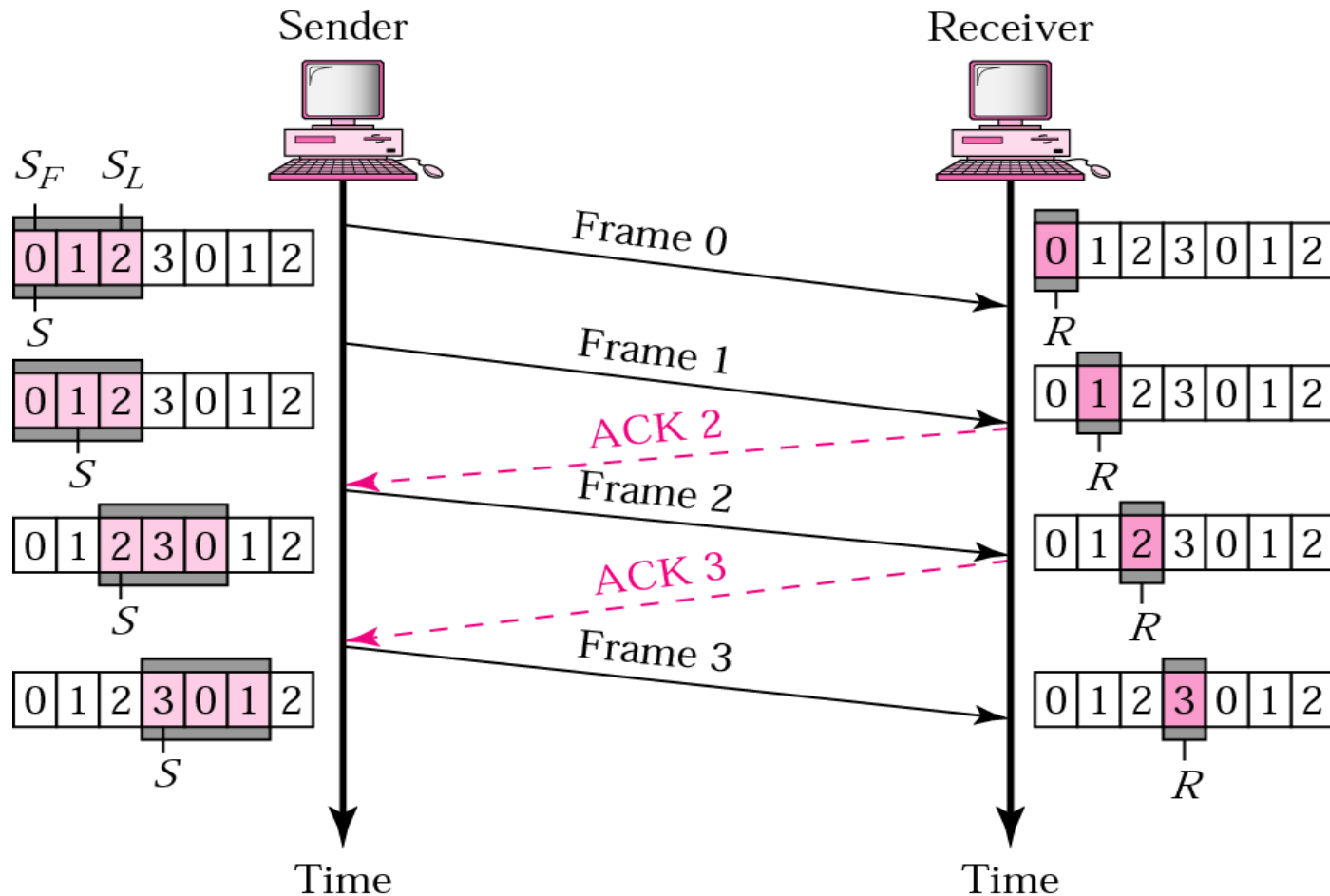


Window Size : 4

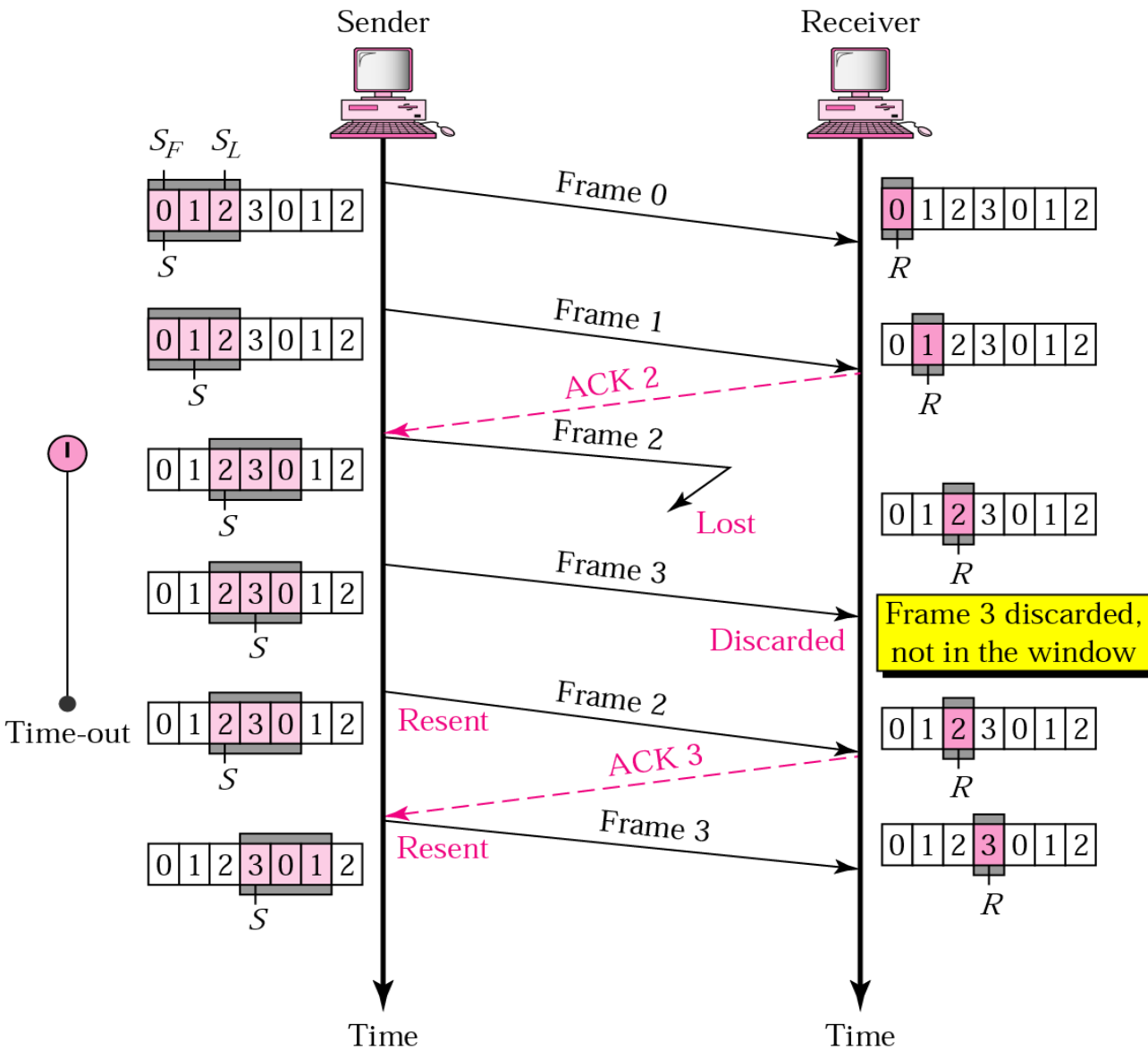


operation

- The sender keeps track of the outstanding frames and updates the variables and windows as the ACKs arrive.



Go Back N ARQ, lost frame



- Frame 2 is lost
- When the receiver receives frame 3, it discards frame 3 as it is expecting frame 2 (according to window).
- After the timer for frame 2 expires at the sender site, the sender sends frame 2 and 3 (go

Go-Back-N ARQ, damaged/lost/delayed ACK

- If an ACK is damaged/lost, we can have two situations:
- If the next ACK arrives before the expiration of any timer, there is no need for retransmission of frames because ACKs are cumulative in this protocol.
- If ACK1, ACK2, and ACK3 are lost, ACK4 covers them if it arrives before the timer expires.
- If ACK4 arrives after time-out, the last frame and all the frames after that are resent.
- Receiver never resends an ACK.
- A delayed ACK also triggers the resending of frames

Selective Repeat ARQ Protocol

- The Go-back-N ARQ protocol works well if it has fewer errors.
- But if there is a lot of error in frame & sending frames again. So, we use Selective Repeat ARQ.
- It stands for Selective Repeat Automatic Repeat Request.
- A sliding window method is used in this data link layer protocol.
- If the receiver receives a corrupt frame, it does not directly discard it.
- It sends a negative acknowledgment to the sender.
- The sender sends that only frame again as soon as on the receiving negative acknowledgment.
- It provide efficient data transmission & fastest recovery.

SELECTIVE REPEAT ARQ

- **Selective Repeat ARQ** is also known as the Selective Repeat Automatic Repeat Request.
- **The go-back-n protocol works well if errors are less**, but if the line is poor it wastes a lot of bandwidth on retransmitted frames

- Selective Repeat attempts to retransmit only those packets that are actually lost (due to errors)
 - Receiver must be able to accept packets out of order.
 - An explicit **NAK** can request retransmission of just one packet.
-

➤ Sender's Windows (Ws) = Receiver's Windows (Wr).

➤ Ws = Wr = 2^{m-1}

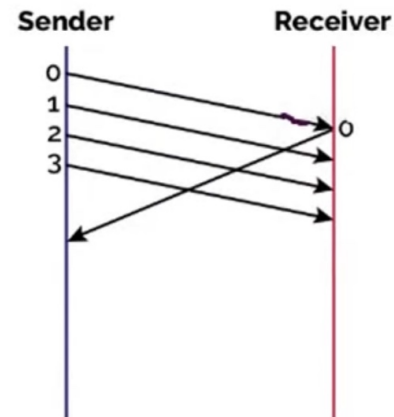
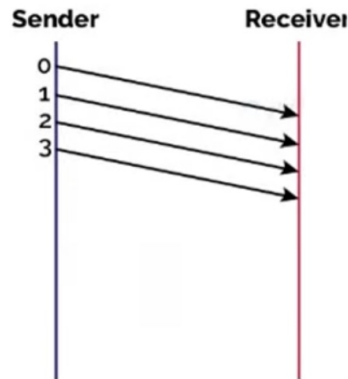
➤ The sender's window must be equal to the receiver's window and the window size should be less than or equal to **half of the sequence number** in the selective repeat protocol. This is to avoid the data packets being recognized incorrectly.

Example of Selective Repeat ARQ

- **Step 1:** Sender will send first four frames to the receiver (0,1,2,3). $N = 4$.
- **Step 2:** Sender expects to get ack from 0th frame.



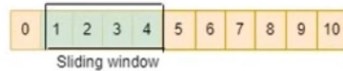
Window Size: 4



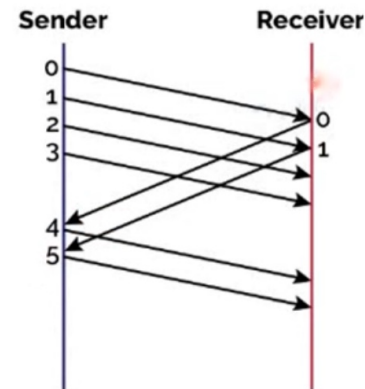
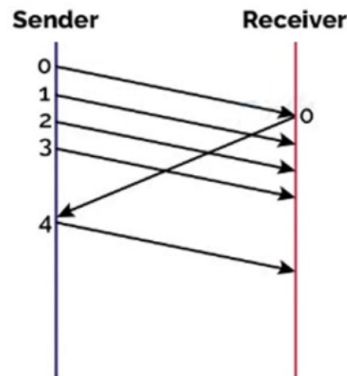
Activate Windows
Go to Settings to activate Windows.

Example of Selective Repeat ARQ

➤ **Step 3:** Sender will send following frame, ➤ **Step 4:** Sender get ack from 1th frame from which is 4. The window slides containing four frames (1,2,3,4).



Window Size: 4



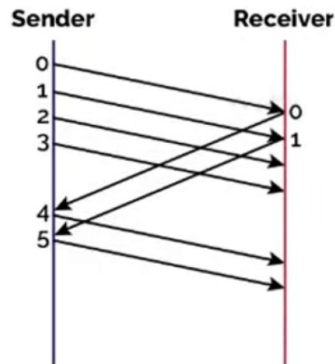
Activate Windows
Go to Settings to activate Windows.

Example of Selective Repeat ARQ

➤ **Step 5:** Sender will then send following frame, which is 5. The window slides containing four frames (2,3,4,5).

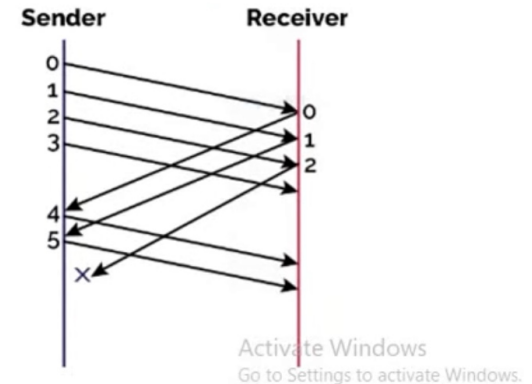


Window Size: 4



➤ **Step 6:** The acknowledgment for frame no. 2 will be sent by receiver. But acknowledgment is lost in network.

Then receiver send Negative acknowledgement (NACK) of frame 2 to the sender.



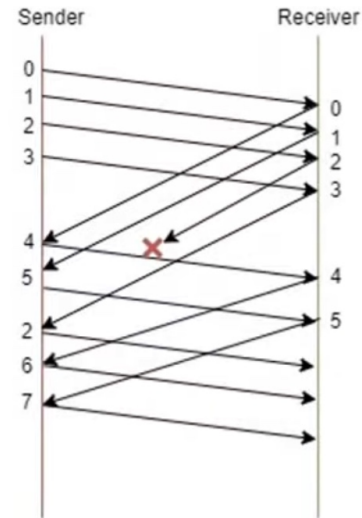
Example of Selective Repeat ARQ

➤ **Step 7:** Sender will then send frame 2 alone to the receiver. As usual they send other frames.



Window Size:

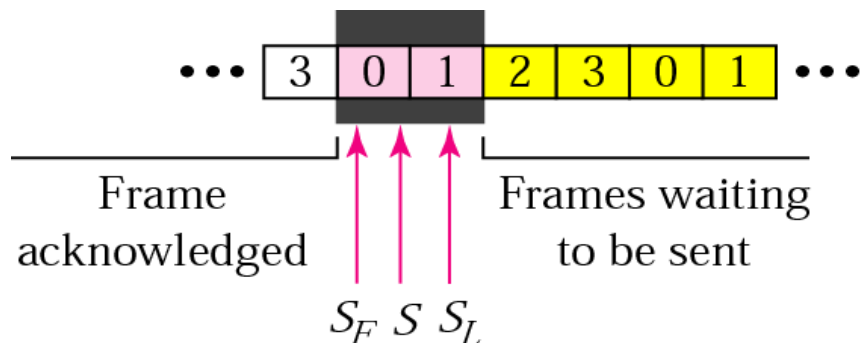
4



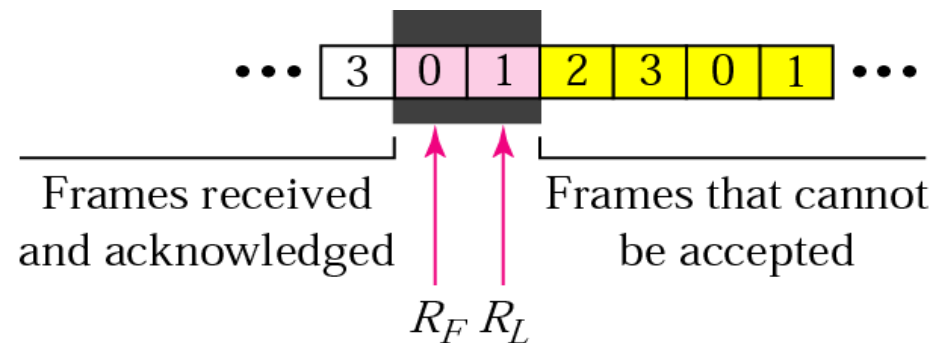
Activate Windows
Go to Settings to activate Windows.

Selective Repeat ARQ, sender and receiver windows

- Go-Back-N ARQ simplifies the process at the receiver site. Receiver only keeps track of only one variable, and there is no need to buffer out-of-order frames, they are simply discarded.
- However, Go-Back-N ARQ protocol is inefficient for noisy link. It bandwidth inefficient and slows down the transmission.
- In Selective Repeat ARQ, only the damaged frame is resent. More bandwidth efficient but more complex processing at receiver.
- It defines a negative ACK (NAK) to report the sequence number of a damaged frame before the timer expires.

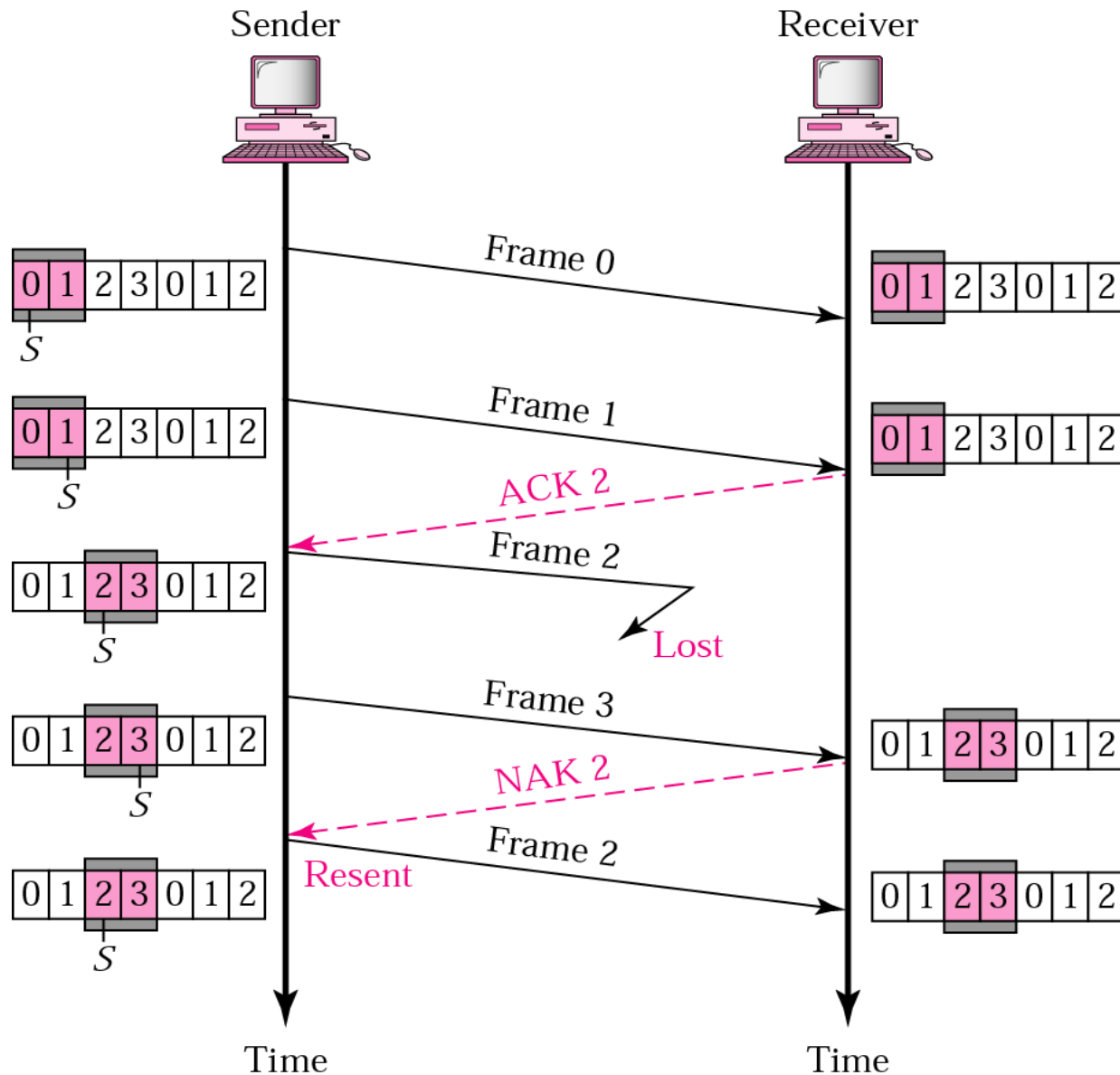


a. Sender window



b. Receiver window

Selective Repeat ARQ, lost frame



Frames 0 and 1 are accepted when received because they are in the range specified by the receiver window. Same for frame 3.

- Receiver sends a NAK2 to show that frame 2 has not been received and then sender

resends only frame 2 and it